

11.1 COCOMO Model Estimation

The Constructive Cost Model (COCOMO) is used to estimate the effort, development time, and team size required for the AgriTrace project. The Basic COCOMO model uses the following formulas:

- Effort (E) = $a \times (KLOC)^b$ person-months
- Development Time (D) = $c \times (E)^d$ months
- People Required (P) = E / D

For a semi-detached software project:

- $a = 3.0, b = 1.12, c = 2.5, d = 0.35$

Estimation: The AgriTrace codebase comprises approximately 8.5 KLOC (thousand lines of code), including TypeScript source files, configuration files, and Firestore security rules.

Table 11.1: COCOMO Effort Estimation

Parameter	Formula	Value
Lines of Code (KLOC)	Measured	8.5
Effort (E)	$3.0 \times (8.5)^{1.12}$	33.2 person-months
Development Time (D)	$2.5 \times (33.2)^{0.35}$	8.3 months
People Required (P)	$33.2 / 8.3$	4.0 persons
Productivity	KLOC / E	0.26 KLOC/PM

The COCOMO estimation suggests approximately 33 person-months of effort distributed across 4 team members over 8 months. Due to the use of high-level frameworks (Next.js, Firebase) and component libraries (Radix UI, Tailwind CSS), the actual development time was compressed to approximately 3 months, demonstrating the productivity gains of modern web development tools.

11.2 Itemized Cost Breakdown

The following table provides a detailed cost estimation for the AgriTrace project:

Note: The majority of the technology stack used in AgriTrace is open-source and free, significantly reducing development costs. Firebase and Vercel offer generous free tiers that are sufficient for development and small-scale production deployments. The open-source nature of the tools also eliminates vendor lock-in, allowing future migration to alternative platforms if needed.

11.3 Resource Allocation

The following table shows the resource allocation across team members: